

C++ Programming Language

Conditional operators/Ternary op.

Syntax: (Cond.) ? exp1 : exp2 ;

(Note: Cond. is true, exp1 is false)

```
num1 = 10;
```

```
num2 = 20;
```

```
void main () {
```

```
int num1 = 10;
```

```
int num2 = 20;
```

```
(num1 > num2) ? print("Num1 is greater than Num2") :  
print("Num2 is greater than Num1");
```

odd/Even

```
void main () {
```

```
int num1 = 12;
```

```
var result = (num1 % 2 == 0) ? "Num1 is Even" :  
"Num1 is odd";
```

```
print(result);
```

```
}
```

— x —

```
  111  
x 001  
---  
  001
```

```
  1001  
x 0001  
---  
  0001
```

```
  1000  
x 0001  
---  
  000
```

```
num1 = 12 { Positive  
           { Zero  
           { Negative
```

```
void main () {
```

```
int num1 = 10;
```

```
var result = (num1 > 0) ? "num1 is Positive  
Number"
```

```
: (num1 < 0) ? "num1 is Negative  
Number"
```

```
: "num1 is zero";
```

```
print(result);
```

```
}
```

00	- 00
01	- 01
02	- 10
03	- 11
04	- 100
05	- 101
06	- 110
07	- 111
08	- 1000
09	- 1001
10	- 1010

Marks <u>(80-100)</u> →	<u>Grade</u> <u>A+</u>	<u>(0-100)</u>
<u>(60-79)</u> →	<u>A</u>	
<u>(40-59)</u> →	<u>P</u>	
40 ↓ <u>(0-39)</u> →	<u>F</u>	

```
void main() {
```

```
    int marks = 67;
```

```
    var result = (marks >= 80 && marks <= 100)
```

↓
R.op
↓
Logical op
↓
R.op

```
        ? "Your Grade is : A+" :
```

```
    (marks >= 60 && marks <= 79)
```

```
        ? "Your Grade is : A" :
```

```
    (marks >= 40 && marks <= 59)
```

```
        ? "Your Grade is : P" :
```

```
    (marks >= 0 && marks <= 39)
```

```
        ? "Your Grade is : F" :
```

```
        "Invalid Input" ;
```

```
    print(result);
```

```
}
```

$\text{num} = \underline{3} \rightarrow \text{"Fizz"}$
3, 6, 9,
 $\underline{5} \rightarrow \text{"Buzz"}$
5, 10, 15, 20
 $\underline{(3 \& 5)} \rightarrow \text{"FizzBuzz"}$
15, 30, 45,
other $\rightarrow \text{"Invalid Input"}$

```

void main () {
    int num1 = 15;
     $\text{result} = (\text{num1} \% 3 == 0 \ \&\& \ \text{num1} \% 5 \neq 0) ? \text{"Fizz"} :$   

 $(\text{num1} \% 5 == 0 \ \&\& \ \text{num1} \% 3 \neq 0) ? \text{"Buzz"} :$   

 $(\text{num1} \% 3 == 0 \ \&\& \ \text{num1} \% 5 == 0) ? \text{"FizzBuzz"} :$   

     $\text{"Invalid Input"};$   

    printf("%s", result);
}

```

1)

Temp = 10;

40 $\uparrow$ $\rightarrow$ Hot ✓

30 - 39 $\rightarrow$ Pleasant ✓

20 $\downarrow$ $\rightarrow$ Cool ✓

2)

Shopping Amt $\geq$ (5000 and above)

$\rightarrow$ Get 20% Discount

(1000 - 4999) $\rightarrow$ Get 10% Discount

(0 - 999) $\rightarrow$ No Discount